

MOROGORO, Tanzania, United Republic ofWMO: **638660**Lat: **6.833S**Lon: **37.650E**Elev: **526**StdP: **95.16**Time Zone: **3.00 (E03)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	16.1	11.8	9.2	23.7	12.7	9.7	23.6	6.1	24.8	5.3	24.6	0.0	240	0.240

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.0	34.2	23.0	33.5	22.9	32.9	22.8	25.2	30.5	24.8	30.0	24.4	29.5	1.9	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.9	20.1	27.4	23.3	19.3	26.6	23.1	19.0	26.3	80.4	30.6	78.5	30.1	76.9	29.6	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	6.2	5.3	13.1	36.1	0.4	1.0	12.8	36.8	12.6	37.4	12.4	38.0	12.1	38.7		
			12.0	27.5	1.1	1.4	11.2	28.5	10.6	29.3	10.0	30.1	9.2	31.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.1	27.4	27.2	26.7	25.2	24.1	22.6	22.2	23.0	24.0	25.5	26.5	27.2
	DBStd	2.15	1.34	1.14	1.26	1.03	0.95	1.17	1.08	0.99	1.05	1.02	1.21	1.20
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5517	538	483	517	457	437	377	377	403	422	479	496	532
	CDD18.3	2476	280	249	259	207	178	127	119	144	172	221	246	273
	CDH23.3	20025	2552	2343	2174	1210	934	779	783	1043	1457	1952	2266	2533
(13)	CDH26.7	7240	1097	1028	879	319	172	118	109	207	467	784	986	1072
(14)	Wind	WSAvg	1.0	2.0	1.6	0.7	0.1	0.1	0.4	0.7	1.3	1.3	1.7	1.8
(15) Precipitation	PrecAvg	773	97	89	111	180	64	15	12	6	11	28	59	108
	PrecMax	1192	219	202	238	317	151	51	66	33	75	102	200	300
	PrecMin	433	11	0	24	88	16	0	0	0	0	0	0	16
	PrecStd	194	66	55	54	58	35	14	18	9	19	29	57	75
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	35.1	34.5	31.9	30.1	29.9	29.6	30.2	32.1	33.4	34.2	34.4
		MCWB	22.7	23.4	24.2	24.1	22.5	20.9	20.1	20.2	21.1	21.9	22.2	23.3
	2%	DB	34.0	34.0	33.2	30.7	29.2	28.9	28.8	29.2	31.1	32.4	33.2	33.5
		MCWB	22.7	23.3	23.7	23.9	22.5	20.8	19.7	19.9	20.8	21.7	22.3	23.0
	5%	DB	33.0	33.0	32.2	29.8	28.5	28.0	28.0	28.8	30.2	31.8	32.8	32.8
		MCWB	22.9	23.1	23.4	23.8	22.4	20.6	19.6	19.7	20.4	21.5	22.3	23.0
	10%	DB	32.0	32.0	31.1	28.8	27.8	27.2	27.1	28.0	29.2	30.8	31.8	31.8
		MCWB	23.0	23.0	23.5	23.5	22.3	20.4	19.5	19.5	20.1	21.3	22.3	23.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	25.3	25.5	25.3	24.2	22.7	22.4	21.8	23.1	24.8	25.5	25.8
		MCDB	31.0	31.2	31.5	29.4	27.5	27.0	27.2	27.1	29.6	30.2	30.4	30.8
	2%	WB	24.6	24.7	25.0	24.7	23.7	22.2	21.3	21.2	21.8	23.2	24.5	24.9
		MCDB	30.5	30.8	30.6	28.8	26.9	26.6	26.2	26.5	28.3	28.6	29.6	30.0
	5%	WB	24.2	24.1	24.5	24.3	23.2	21.7	20.7	20.6	21.2	22.5	23.7	24.2
		MCDB	30.0	30.3	29.8	28.1	26.5	26.1	25.6	26.0	27.6	28.1	29.0	29.6
	10%	WB	23.7	23.7	24.1	23.9	22.8	21.2	20.1	20.2	20.7	22.0	23.1	23.7
		MCDB	29.2	29.5	28.9	27.5	26.1	25.5	25.1	25.6	26.9	27.8	28.3	29.0
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.0	9.5	7.8	8.1	10.5	10.9	10.9	11.9	11.3	10.7	9.9
		MCDBR	10.9	11.4	11.0	9.2	9.3	11.2	12.0	11.9	12.8	12.7	12.2	11.1
	5% WB	MCWBR	2.9	3.2	3.2	3.7	3.9	4.7	4.9	3.9	4.0	3.7	3.1	3.0
		MCWBR	9.7	10.2	9.7	8.0	7.6	9.9	10.5	10.7	11.9	11.1	10.3	9.8
(40) Clear-Sky Solar Irradiance	taub	taub	0.446	0.450	0.409	0.364	0.362	0.368	0.395	0.406	0.399	0.392	0.386	0.412
		taud	2.318	2.310	2.441	2.598	2.557	2.486	2.364	2.332	2.355	2.394	2.453	2.422
	Ebn at Noon	Ebn at Noon	895	890	914	929	904	883	863	877	909	932	945	923
		Edh at Noon	137	138	120	98	98	102	117	126	128	126	119	123
(44) All-Sky Solar Radiation	RadAvg	RadAvg	5.81	5.93	5.54	4.82	4.54	4.53	4.50	4.63	5.18	5.52	5.76	5.84
	RadStd	RadStd	0.45	0.35	0.36	0.31	0.34	0.21	0.23	0.21	0.31	0.32	0.30	0.32

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page